CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

ENTHUSIAST ADVANCE COURSE

PHASE : MEF-I (B)

TARGET : PRE MEDICAL 2024

Test Type : **MINOR** Test Pattern : **NEET (UG)**

TEST DATE : 16-07-2023

ANSWER KEY

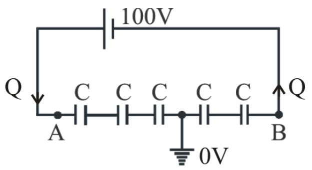
Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	4	2	3	2	2	2	2	4	1	2	2	3	2	1	4	3	1	2	1	1	3	3	3	3	2	4	4	4	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	1	4	2	2	1	3	2	3	2	2	3	2	2	1	4	2	2	4	3	3	3	3	4	4	3	1	3	3	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	2	1	1	3	4	3	2	3	3	1	4	4	4	4	3	4	2	3	4	2	3	3	4	2	4	3	2	4	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	2	1	3	4	3	3	4	1	3	1	3	2	2	1	2	1	4	4	1	1	3	2	4	3	4	2	1	1	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	3	4	3	1	1	3	1	1	3	2	4	3	2	4	1	1	1	2	3	3	3	3	3	3	3	1	1	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	4	4	1	4	4	2	3	2	4	1	3	2	4	4	4	1	4	3	2	3	4	4	4	3	2	4	3	4	1
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	1	2	2	1	1	1	1	1	4	4	3	4	3	3	1	1	4	4	2	2										

HINT – SHEET

SUBJECT : PHYSICS

SECTION - A

1. **Ans (3)**



$Q = C_{eq} V$

$$Q = \frac{C}{5} \times 100 = 20\ C$$

$$V_A - 0 = \frac{Q}{C/3} \qquad 0 - V_B = \frac{Q}{C/2}$$

$$V_A = \frac{20C \times 3}{C} V \qquad -V_B = \frac{20C \times 2}{C}$$

$$= 60\ V \qquad V_B = -40\ V$$

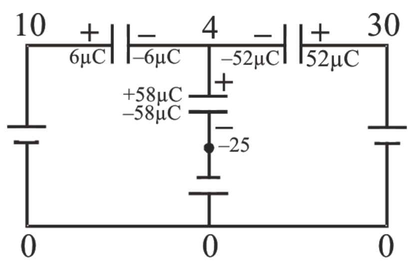
2. **Ans (4)**

Let potential at A is 0, so at D it is 30V, at F it is 10V and at point G potential is –25V and let potential at E is x. Now apply kirchhoff's Ist law at point E. (Total charge of all the plates connected to 'E' must be same as before i.e. 0)

$\therefore\ (x - 10) + (x - 30)2 + (x + 25)2 = 0$

$5x = 20$

$x = 4V$



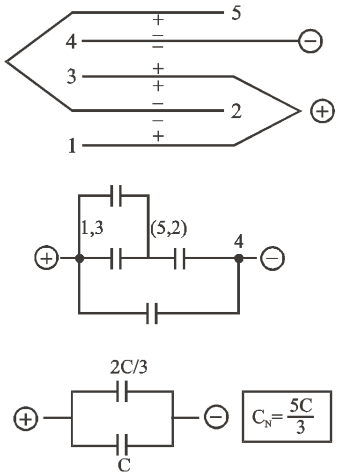
Final charges :

$Q_{2\mu F} = (30 - 4)2 = 52\mu C$

$Q_{1\mu F} = (10 - 4) = 6\mu C$

$Q_{2\mu F} = (4 - (-15))2 = 58\mu C$

3. Ans (2)



4. Ans (3)

$C_{eq} = KC$, $q' = C_{eq}V = KCV = Kq$

$U = \frac{1}{2}CV^2$

$U' = \frac{1}{2}KC.V^2 = KU$

$\Delta U = U' - U = \frac{1}{2}CV^2(K-1)$

5. Ans (2)

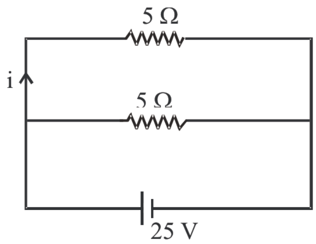
$W_{batt} = (q)V$

$= 10^{-6} \times 300$

$= 3 \times 10^{-4}$ J

6. Ans (2)

Given circuit can be reduced to

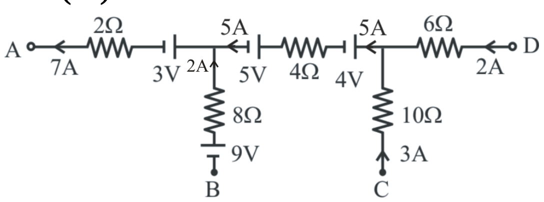


Therefore $i = \frac{25V}{5\Omega} = 5A$

7. Ans (2)

In zero deflection condition, potentiometer draws no current.

8. Ans (2)



$V_A - V_C = -3(10) - 4 - 5(4) - 5 - 3 - 7(2)$

$= -30 - 4 - 20 - 5 - 3 - 14$

$= -76$

9. Ans (4)

here

$V_A = V_C$

$\therefore$ no current flows between A and C

(i.e. through resistance of 12 Ω)

10. Ans (1)

$V = IR = x\ell$

$R \propto \ell$

$\frac{R_1}{R_2} = \frac{\ell_1}{\ell_2} \Rightarrow \frac{5}{4} = \frac{40}{x}$

$x = \frac{40 \times 4}{5} = 32$ cm

16. Ans (4)

$B = \frac{\mu_0 Ni}{2R}$

$= \frac{4\pi \times 10^{-7} \times 1000 \times 1}{2 \times 62.8 \times 10^{-2}}$

$= \frac{4 \times 3.14 \times 10^{-7} \times 10^3}{2 \times 62.8 \times 10^{-2}} = 10^{-3}$ T

19. Ans (2)

Tension in the string

$T = M(g - a) = M(g - \frac{g}{2}) = \frac{Mg}{2}$

$W = \text{Force} \times \text{Displacement} = -M\left(\frac{g}{2}\right)h$

$-$ve sign is because tension is in upward direction and displacement is in downward direction.

20. Ans (1)

$\because KE = \frac{p^2}{2m}$

$\frac{\Delta KE}{KE} = 2\left(\frac{\Delta p}{p}\right)$

$\Rightarrow \left(\frac{\Delta p}{p}\right) = \frac{1}{2}\left(\frac{\Delta KE}{KE}\right)$

% change in momentum $= \frac{1}{2}(3\%)$

$= 1.5\%$

21. Ans (1)

Weight of overhanging part of the chain $= \frac{mg}{6}$

The weight acts at the centre of gravity of the overhanging part, i.e., $\frac{1}{12}$ below the surface of the table.

Gain in potential energy $= \frac{mg}{6} \times \frac{1}{12} = \frac{mgl}{72}$

$\therefore$ Work done = gain in potential energy $= \frac{mgl}{72}$.

22. **Ans (3)**

N = 200 Newtons.

h = 20 × 0.25 m

= 5m

$\therefore$ W = N.h

= 200 × 5 J

$\Rightarrow$ W = 1000 J

23. **Ans (3)**

$$W = \int \vec{F}.\vec{ds}$$

$$= \int F.\ dx$$

$$= \int_{x=0}^{5} 8x.\ dx$$

$$= \left.\frac{8x^2}{2}\right|_0^5$$

= 4 × 25

$\Rightarrow \boxed{W = 100J}$

24. **Ans (3)**

$$U_i = \frac{1}{2}kx_1^2$$

$$\Rightarrow 100 = \frac{1}{2}k\left(\frac{2}{100}\right)^2$$

$$\Rightarrow \boxed{k = 5 \times 10^5 N/m}$$

$$\therefore U_f = \frac{1}{2}k\,x_2^2 \quad \{x_2 = (2+2)cm = 4cm\}$$

$$= \frac{1}{2} \times 5 \times 10^5 \times \left(\frac{4}{100}\right)^2$$

= 400J

$\therefore \Delta U = U_f - U_i$

= 400J – 100J

$\Rightarrow \boxed{\Delta U = 300J}$

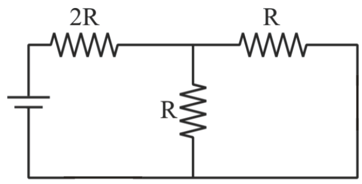
29. **Ans (4)**

Phase conversion is isothermal process

SECTION - B

36. **Ans (1)**

Just after closed the key, this circuit consider as.



$$C_{eq} = 2R + R/2 = \frac{5R}{2}$$

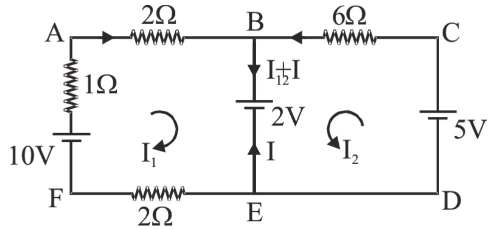
$$\therefore \text{Current} = \frac{V}{R_{eq}} = \frac{2V}{5R}$$

37. **Ans (3)**

$$C_R = C_1 + C_2 = \frac{k_1\varepsilon_0 A_1}{d} + \frac{k_2\varepsilon_0 A_2}{d}$$

$$\frac{2 \times \varepsilon_0 \frac{A}{2}}{d} + \frac{4 \times \varepsilon_0 \frac{A}{2}}{d} = 2 \times \frac{10}{2} + 4 \times \frac{10}{2} = 30\ \mu F$$

38. **Ans (2)**



in loop ABEFA

$$I_1 = \frac{8}{5} A = 1.6\ A$$

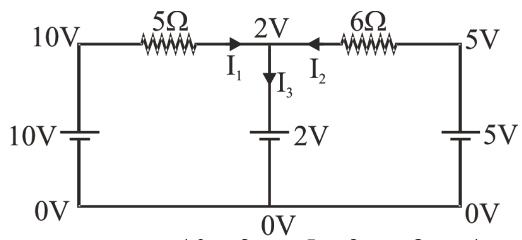
in loop BCDEB

$$I_1 = \frac{3}{6} = 0.5\ A$$

I = –(I1 + I2)

I = –2.1 A

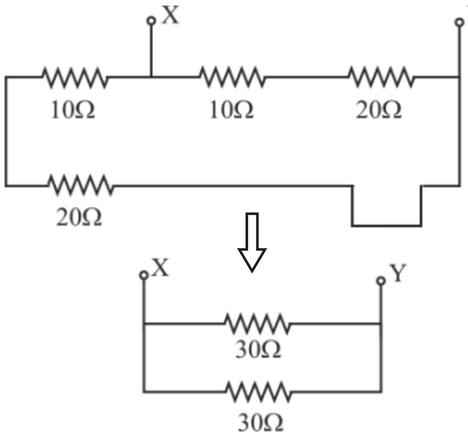
OR



$$I_3 = I_2 + I_2 = \frac{10-2}{5} + \frac{5-2}{6} = \frac{8}{5} + \frac{1}{2} = 2.1A$$

So, according to give arrow in question $I_3 = -2.1A$

39. Ans (3)



So, $R_{XY} = 15\ \Omega$

40. Ans (2)

By given circuit.

$V_x - 10 + 5, - 10 = 0$

$V_x = 15$ V

42. Ans (3)

Let the coordinate at which magnetic field is zero is given by (x, y)

$\therefore \quad B_1 = B_2$

$\frac{\mu_0 3I}{2\pi x} = \frac{\mu_0 I}{2\pi y}$

$3y = x$

46. Ans (4)

$$W = \int_{x_1}^{x_2} F.dx = \int_0^5 (7 - 2x + 3x^2)dx$$

$$= \left| 7x - x^2 + x^3 \right|_0^5 = 7(5) - (5)^2 + (5)^3$$

$$= 35 - 25 + 125 = 135 \text{ J}$$

47. Ans (2)

$$P = \vec{F}.\vec{V} \qquad \begin{Bmatrix} \theta = 0 \\ \cos\theta = 1 \end{Bmatrix}$$

$= F.V$

$= 4500 \times 2$ W

$= 9000$ W

$\Rightarrow \boxed{P = 9\text{kW}}$

49. Ans (4)

Let θ be the final common temperature. Further, let s_c and s_h be the average heat capacities of the cold and hot (initially) bodies respectively (where $s_c < s_h$ given)

From, principle of calorimetry,

heat lost = heat gaine

$s_h(80°C - \theta) = s_c\theta$

$\therefore \quad \frac{s_h}{(s_h + s_c)} \times 80°C = \frac{80°C}{\left(1 + \frac{s_c}{s_h}\right)}$

$\because \quad s_c / s_h < 1$

$\therefore \quad 1 + s_c / s_h < 2$

$\therefore \quad \theta > \theta > \frac{80°C}{2}$ or $\theta > 40C$

OR

Body at 80°C has more heat capacity then body at 0°C so final temperature must be greater than 40°C.

SUBJECT : CHEMISTRY

SECTION-A

57. Ans (1)

$r_{Mg} > r_{Al}$

58. Ans (3)

$\Delta H_{eg} \quad Cl > F$

61. Ans (2)

x − 1 + 0 = +2, covalency = 1 + 5 = 6

x = +3

62. Ans (2)

Covalent bond are directional bond.

69. Ans (3)

Metal above the Cu in ECS can displace Cu^{+2} from $CuSO_4$. Ag is below Cu. Hence less reactive than Cu.

70. Ans (3)

For 1 equivalent substance 1F charge is required.

71. Ans (1)

$\lambda_m^\infty(NH_4OH) = \lambda_m^\infty(NH_4Cl) + \lambda_m^\infty(NaOH) - \lambda_m^\infty(NaCl)$

72. **Ans (4)**

$$\lambda_{eq} = \frac{\lambda m}{V.F.}$$

Here, V.F. = 6

$$\lambda_{eq} = \frac{\lambda m}{6}$$

SECTION-B

88. **Ans (2)**
except He ($1s^2$)

89. **Ans (4)**
for same n value pinetration effect is applicable.

90. **Ans (1)**
IE_1 of Al is in between Na - Mg values.

95. **Ans (4)**
$Pb + PbO_2 + 2H_2SO_4 \longrightarrow 2PbSO_4 + 2H_2O$

SUBJECT : BIOLOGY-I

SECTION-A

102. **Ans (3)**
NCERT Pg. # 181

103. **Ans (2)**
NCERT Pg. # 187

104. **Ans (2)**
NCERT Pg. # 119 (2nd Para), 120 (Point ix), Page 121 (3rd, 4th Para), Page 122 (Last Para)

105. **Ans (1)**
NCERT Pg. # 114

110. **Ans (1)**
NCERT XII Pg. # 198

113. **Ans (2)**
NCERT-XII, Pg.#112

114. **Ans (4)**
NCERT-XII Pg. # 113

115. **Ans (3)**
NCERT-XII Pg. # 115

116. **Ans (4)**
NCERT-XII Pg. # 116

117. **Ans (2)**
NCERT-XII Pg. # 118

118. **Ans (1)**
NCERT-XII Pg. # 120

119. **Ans (1)**
NCERT-XII Pg. # 121

120. **Ans (4)**
NCERT-XII Pg. # 121

121. **Ans (1)**
NCERT-XII Pg. # 164

122. **Ans (1)**
NCERT-XII Pg. # 165

123. **Ans (3)**
NCERT-XII Pg. # 166,168

124. **Ans (4)**
NCERT-XII Pg. # 174

127. **Ans (1)**

129. **Ans (1)**
NCERT-XII Pg. # 53 para-1

130. **Ans (1)**
NCERT–XII, Pg. # 52, Fig.-3.11(g)

135. **Ans (2)**

SECTION-B

136. **Ans (4)**
NCERT (XII) Pg. # 181, 183

138. **Ans (1)**
NCERT Pg. # 117

139. **Ans (1)**
NCERT Page No. 132

143. **Ans (3)**
NCERT (XII) Pg. # 201, para-11.3

144. **Ans (3)**
NCERT-XII Pg. # 117

145. Ans (3)

NCERT-XII Pg. # 122

SUBJECT : BIOLOGY-II

SECTION-A

152. Ans (4)

NCERT Pg. # 58

161. Ans (1)

NCERT Pg.# 60

162. Ans (3)

NCERT Pg. # 260 (Fig. 15.1)

164. Ans (4)

NCERT (XIIth) Pg. # 265

169. Ans (3)

NCERT (XII) Pg. # 277

170. Ans (2)

NCERT (XII) Pg. #184, 185

SECTION-B

186. Ans (1)

Both Statements are correct.

189. Ans (4)

NCERT XII Pg # 60

192. Ans (4)

NCERT (XII) Pg. # 281

193. Ans (3)

NCERT (XIIth) Pg. # 273